

Mohamed Khalil ZRELLY

Generative AI Engineer | Agentic Systems | RAG Pipelines | LLM Architectures

Email: mohamed.khalil.zrelly@gmail.com | Phone: +33 7 58 21 58 56 | Location: Le Bourget, France

LinkedIn: linkedin.com/in/mohamed-khalil-zrelly | GitHub: github.com/makiz-code | Portfolio: makiz-code.github.io/MKirell

Professional Summary

Generative AI Engineer Apprentice at Crédit Agricole, specializing in agentic systems, RAG pipelines, and LLM architectures. Engineering student at Sup Galilée — Sorbonne Paris North University, with a background in machine learning, deep learning, and natural language processing.

Work Experience

Crédit Agricole Personal Finance & Mobility France
Generative AI Engineer Apprentice September 2025 — Present · 10 months

- Developed a declarative, config-driven **graph workflow orchestration** microservice, enabling dynamic workflow composition across a scalable and maintainable AI ecosystem.
- Built an automated **Red Teaming** framework for AI workflows, probing **LLM behaviour** across guardrails and targeted adversarial scenarios to strengthen model resilience.

Agentic Workflows · Advanced RAG · AWS Bedrock · AI Guardrails · LLM Observability · Python/FastAPI · Amazon DynamoDB

VERMEG for Banking & Insurance Software Tunisia
Natural Language Processing Intern February 2024 — June 2024 · 5 months

- Implemented a system that automates **email classification** and information extraction from attachments, using **BERT** and **OCR** to optimise efficiency and processing time.
- Developed a web application for **BERT Fine-Tuning**, email management, and real-time data analysis and visualisation, to enhance user interaction and performance monitoring.

Hugging Face Transformers · Fine-Tuning · Named Entity Recognition · PyTorch · Tesseract OCR · Python/Flask · React/Redux · MongoDB

Sofiatech Tunisia
Machine Learning Intern June 2023 — August 2023 · 3 months

- Evaluated multiple **deep reinforcement learning** algorithms from **Stable Baselines 3** across various **Gymnasium** environments to assess agent performance and training efficiency.
- Built a **2D grid game** in **Pygame** where a **Deep Q-Learning** agent learns to navigate a dynamic environment, applying the best-performing algorithm from the benchmarking phase.

Reinforcement Learning · Deep Q-Learning · stable-baselines3 · TensorFlow/Keras · Gymnasium · NumPy · Pandas · Matplotlib

Featured Projects

CVision | RAG · CV Recommendation System June 2025 — August 2025

Created an intelligent **CV recommendation system** based on job description using **Retrieval-Augmented Generation (RAG)**, to optimise the selection and automatic ranking of the most relevant profiles.

Advanced RAG · Re-Ranking · LangChain · Qwen3-VL · Weaviate · Python/Streamlit

Twixel | Multi-Agent · Social Listening System In Development — 2026

Designing an intelligent **multi-agent system** for automated social listening on X (Twitter), aimed at identifying emerging trends in real time and generating actionable marketing campaign ideas at scale.

Multi-Agent Systems · LangGraph · MCP/A2A · GPT-OSS · Milvus · Python/FastAPI · Vue.js

Technical Skills

Generative AI & Agentic Systems	LangChain, LangGraph, AWS SageMaker, AWS Bedrock
ML / DL / RL	PyTorch, TensorFlow, Keras, Scikit-Learn, Stable Baselines 3, Gymnasium
NLP & LLM Architectures	Hugging Face Transformers, spaCy, Fine-Tuning, RAG
Data Analysis & Big Data	Pandas, NumPy, Matplotlib, Apache Airflow, PySpark
SQL & NoSQL Databases	PostgreSQL, MongoDB, Amazon DynamoDB, MySQL, Oracle
Web Full Stack & REST APIs	React, Vue.js, FastAPI, Flask, Express.js, Laravel
DevOps & Cloud Infrastructure	AWS, Docker, Kubernetes, Terraform, GitHub Actions
Programming Languages	Python, JavaScript/TypeScript, Java, C/C++, PHP, PL/SQL

Education

Sup Galilée – Sorbonne Paris North University Villetaneuse, France
Engineering Degree in Computer Science 2024 — 2027 (Expected)

ISI Ariana – University of Tunis El Manar Ariana, Tunisia
Bachelor of Science in Computer Science 2021 — 2024 · High Distinction

Certifications & Languages

Microsoft Azure	Azure AI Fundamentals (AI-900) — Jan 2024 · Azure Data Fundamentals (DP-900) — Jan 2024
Udemy	Spark and Python for Big Data with PySpark — Mar 2024 · NLP with Python — Feb 2024
Spoken Languages	Arabic (Native) · French B2 — TCF · English B2 — TOEIC